

- Challenging problems: 1-4,
- Input a sequence of N numbers. List the negative numbers in the sequence.
- Input a sequence of N integers. Count how many are even and calculate the sum of the odd numbers.
- Input a sequence of N numbers and a number K . Count how many elements in the sequence are less than K .
- Input a sequence of N numbers. Find the value(s) that appear most frequently in the sequence.

1. Write a function `findMax` that takes three integers as parameters and returns the largest number among them.
2. Write a function `findMin` that takes three integers as parameters and returns the smallest number among them.
3. Write a function `isPrime` that checks if a positive integer n is a prime number.
4. Write a function `sumDivisors` to calculate the sum of all positive divisors of a positive integer n .
5. Write a function `sumDigits` to calculate the sum of the digits of an integer $n > 0$.
6. Write a function that takes a non-negative integer n as a parameter and finds the smallest and largest digits of n .
7. Write a function that takes an integer $N > 1$ and a real number x as parameters and returns the sum:

$$e^x \approx S = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$$



- Given a square matrix, print the elements on the main diagonal and the elements on the secondary diagonal.
- Given a square matrix, swap two columns.
- Given a square matrix, swap two columns.
- Given a $M \times N$ matrix. Find the value(s) that appear most frequently in the sequence.
- Given a vector of N numbers, sort these numbers in ascending order.
- Given a vector of N numbers, reverse the elements in the vector.
- Given a vector of N integers, print the set containing only distinct numbers (removing duplicates).

1. Input an array of N numbers and sort them in descending order.
2. Input an array of N integers, print out the elements that are divisible by the first element.
3. Input an array of N elements, find the indices of the elements with the minimum value.
4. Programming Challenges: 19
5. Given a square matrix, write two functions to swap two rows and two columns, respectively.
6. Given a vector of N integers, print the set containing only distinct numbers (removing duplicates).